Worksheet — Free Fall Simulation

Use with *Simulation - Free fall.html* · A.1 Kinematics: free fall, suvat, *h* against t^2 → *g*; goes with the class ball-drop experiment (stopwatch)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/10-free-fall** in a browser (offline: *Simulation - Free fall.html*). Set the **height h**, the **launch velocity v_0** (0 = drop, + = throw up), the **field**, ball A and an optional ball B. ▶ **Drop** runs it; the **strobe** shows the ball every 0.1 s.

Part 1 before the lab (≈ 15 min) · **Part 2** during the ball-drop lab (≈ 15 min) · **Part 3** homework (≈ 30 min).

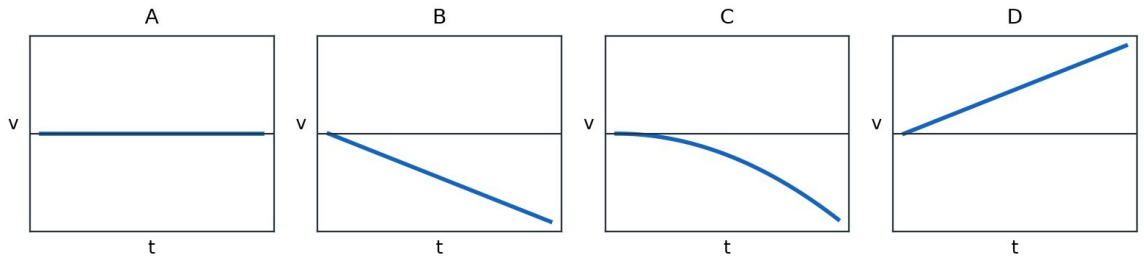
Use g = 9.81 m s^{-2} and **up is positive**: a falling ball has a negative velocity and a = −9.81 m s^{-2}.

Scan to open

Part 1 — Pre-lab: predict, then test

Write your prediction **before** you run the simulation. Then test it and explain what you saw.

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Steel ball (64 g) and ping-pong ball (2.7 g) dropped together from 1.00 m in a **vacuum**. Which lands first?			
2. Air resistance **on**: steel ball and paper cupcake case dropped from 2.00 m. What happens?			
3. The drop height is doubled from 1.00 m to 2.00 m. By what factor does the fall time change?			
4. Which *v*–*t* graph (A–D below) shows a ball dropped from rest? (up positive)			



Q4 options (up is positive)

5. Calculate the time for a ball to fall 1.00 m from rest. Then check it with the simulation (light gate).

t = _____ s simulation: _____ s

Part 2 — During the lab: compare with the ball drop

Class experiment. Drop a ball from each height and time the fall with a stopwatch. Do 3 drops per height.

h / m	t_1 / s	t_2 / s	t_3 / s	mean t / s	t^2 / s^2
0.50					
1.00					
1.50					
2.00					

Simulation. Timing = **Stopwatch**. Do the same drops (3 per height) and press ● **Record** each time.

h / m	t_1 / s	t_2 / s	t_3 / s	mean t / s	t^2 / s^2
0.50					
1.00					
1.50					
2.00					

1. Calculate the true fall time from 0.50 m. How does it compare with your stopwatch times?

2. What is the main cause of the spread in your stopwatch times? Is it a random or a systematic uncertainty?

3. g from the simulation's best-fit line (Data tab, g = 2 × gradient): g = _____ m s^{-2}

4. g from the class data (plot h against t^2 in Part 3 or on graph paper): g = _____ m s^{-2}

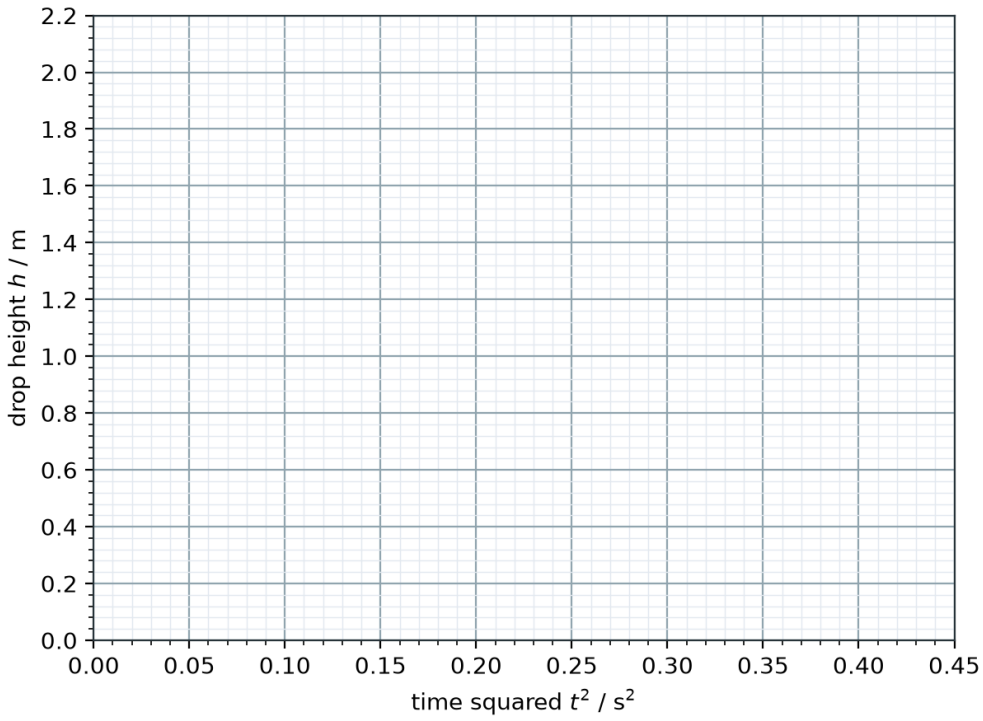
5. Percentage difference from 9.81 m s^{-2}, and one improvement to the experiment:

Part 3 — Homework: analyse

Timing = **Real data ±2 %**, steel ball. Record one drop from each height.

h / m	*t* / s	t^2 / s^2
0.40		
0.80		
1.20		
1.60		
2.00		

1–2. Plot h (y-axis) against t^2 (x-axis). Draw the best-fit line through the origin.



3. Gradient = _____ m s^{-2} ⇒ g = 2 × gradient = _____ m s^{-2}

Explain why the gradient of *h* against t^2 is $g/2$:

4. A ball is dropped from rest from 1.80 m. Calculate the time it takes to reach the floor.

5. Calculate the speed of the ball just before it hits the floor.

6. A ball is thrown straight up at 8.0 m s^{-1}. How high does it rise above the hand?

7. How long after the throw is the ball back at the level of the hand?

8. At the very top of the throw, what are the velocity and the acceleration? Explain.

9. An astronaut drops a hammer from 1.80 m on the Moon (g = 1.62 m s^{-2}). Calculate the fall time and compare it with Q4.